

Rupendran Manigandan

Date of birth: 18/03/2006 | Gender: Male | Nationality: Indian | Phone: (+91) 9800089902 (Mobile) | Email address: rupendran2006@gmail.com | Website: https://github.com/RUPEN-18 | Address: No 57, Annai Bhavani St, Nassarat Pettai, Little Kanchipuram, 631501, Kanchipuram, India (Home)

About me

Artificial Intelligence and Robotics engineer pursuing B.E.(CGPA: 7.55/10) with hands-on expertise in deep learning, computer vision, and embedded autonomous systems. Experienced in developing industrial visual inspection models (PyTorch, YOLO) and custom networking solutions during a full-time engineering internship at Larsen & Toubro. Winner of 2nd Prize at the TSSIFThailand International Expo, with 1 industrial patent application under evaluation (Wireless Miracast using TCP/IP) and author of a research paper on Autonomous Robotic Assistant(AURA)

Education & Training

Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence and Robotics) | Sathyabama Institute of Science and Technology | 07/08/2023 - Current | Chennai, India

Academic Status: CGPA 7.55 / 10.0 (Semesters 1–6)

• **Math & Theoretical CS:** Calculus & Numerical Methods, Matrices & Linear Algebra, Discrete Structures, Probability & Statistics, Optimization Techniques, Data Structures & Algorithms (DSA), Theory of Computation, Computer Architecture.

• **Systems & Software Engineering:** Operating Systems (Unix), Database Management Systems (DBMS), Data Communication & Computer Networks, Software Engineering Principles, UNIX Shell Scripting.

• **AI, ML & Computer Vision:** Foundations of AI, Machine Learning for Robotics, Computational Intelligence, Deep Learning & Neural Networks (MATLAB), Computer Vision, Big Data Analytics for AI.

• **Robotics & Embedded Systems:** Fundamentals of Robotics, Robot Actuators, Sensors & Intelligent Instrumentation, Software Robot Design, Robotic Perception, Robot Programming, Mechanical Systems Engineering.

• **Languages & Engineering Tools:** Python, C Programming, SQL, MATLAB, Engineering CAD.

Field(s) of study: Software and applications development and analysis / Robotics / AI | **Final grade:** 7.55 / 10.0 CGPA (Up to Semester 6) | **Level in EQF:** 6 | **National classification:** 4-Year Professional Engineering Undergraduate Degree (AICTE / UGC Recognized) | **Type of credits:** Choice Based Credit System (CBCS) | **Number of credits:** 161 Total Credits (131 Completed up to Semester 6) | **Thesis:** GEO X: Autonomous Path-Planning Mobile Manipulator for Hazardous Environments | **Website:** https://www.sathyabama.ac.in/

Senior Secondary School Certificate (Class 12 - CBSE) | Maharishi International Residential School (MIRS) | 01/04/2022 - 10/03/2023 | Sunguvarchatram, India

Field(s) of study: Higher Secondary Education in Mathematics, Physics, Chemistry, and Computer Science | **Final grade:** 61% | **Level in EQF:** 4 | **Website:** https://www.maharishiirschennai.com/

Secondary School Certificate (Class 10 - CBSE) | Maharishi International Residential School (MIRS) | 01/04/2020 - 31/03/2021 | Sunguvarchatram, India

Field(s) of study: General Secondary Education covering Science, Mathematics, English, and Social Sciences. | **Final grade:** 71.4 % | **Level in EQF:** 3 | **Website:** https://www.maharishiirschennai.com/

Work experience

Artificial Intelligence and Software Engineering Intern | Larsen & Toubro | 15/09/2026 - 15/10/2026 | Chennai, India

Schedule: Full-time Onsite (45 hours/week)

• **AI Weld Defect Detection:** Built a custom YOLO visual inspection pipeline trained on 7,000+ images to detect and grade welding defect severity.

• **Wireless Miracast (TCP/IP):** Engineered a low-latency 1080p@60fps screen-casting protocol to Android devices without a router or native Miracast hardware *(Patent Application Under Evaluation)*.

• **Network Management (AutoShutdown Nova):** Developed a client-server LAN administration platform for real-time monitoring, scheduled power-downs, and central group control.

• **AI PPE Safety Monitoring:** Programmed a real-time YOLO computer vision system for worker counting and PPE compliance verification (helmets and high-visibility jackets).

Robotics Trainee | Industrial IoT Workshop | 23/03/2026 - 01/04/2026 | Chennai, India

Duration: 1-Week Intensive Workshop (37.5 Hours)

• Assembled 3D-printed InMoov humanoid robot head, integrating mechanical joints and wiring.
• Programmed ESP32 microcontroller using PWM signals for multi-axis servo control and relay power circuits.
• Conducted a 2-day live hardware defense showcasing embedded control and biomechanical head architecture.

Skills

AI & Computer Vision | PyTorch | Ultralytics YOLO | OpenCV | Deep Learning | Object Detection & Classification | Image Processing & Segmentation

Robotics & Embedded Systems | SLAM Navigation | Arduino | Raspberry Pi | ESP32 | Robotic Kinematics | Sensor Fusion

Software Development & Networking | Python | C++ | MATLAB | SQL | TCP/IP Networking | Git & GitHub | Socket Programming

Simulations & Tools | Unreal Engine | CAD | Circuits & Power Relays

Communication and Collaboration Skills | Technical Communication | Project Coordination | Cross-Functional Collaboration

Projects

GEO X – Autonomous Mobile Manipulator | 21/08/2025 - 10/08/2026

• Engineered an autonomous mobile manipulator using LIDAR-based SLAM for spatial mapping, path planning, and obstacle avoidance in hazardous environments.
• Integrated sensor fusion, robotic arm kinematic control, and embedded microcontrollers for remote teleoperation and autonomous payload handling.
• Won 2nd Prize at the TSSIF Thailand International Expo for hardware engineering and autonomous systems design.

AURA – Autonomous Robotic Assistant | 20/08/2026 - Current

Developed an indoor service robot integrating the multimodal Gemini AI API to enable natural conversational interaction, real-time visual scene analysis, and autonomous task execution.

Authored a research paper detailing multimodal AI architectures for service robotics *(In Preparation / Under Review)*.

Language Skills

Mother tongue(s): **Tamil**

	Understanding		Speaking		Writing
	Listening	Reading	Spoken production	Spoken interaction	
English	C2	C2	C1	C2	C2
Hindi	C2	B2	B2	C1	B2

Hobbies and Interests

Badminton, Gardening, Strength & Fitness Training.